4.6.4 Reset configuration word (RCW)

The chip uses an external memory interface to import a subset of the reset configuration information from a memory device during reset.

Such information is called reset configuration word (RCW) data.

The pre-boot loader (PBL) loads RCW data from a non-volatile memory device interface, as specified by the RCW source configuration inputs (-see Reset Configuration Word Source for more information). See Pre-Boot Loader for details on the operation of the PBL. Note that this approach does not completely remove the necessity for at least a few power-on reset (POR) configuration signals. As noted, POR config signals are used to control RCW source information in addition to other low-level system configuration.

The logic involved is clocked directly from SYSCLK since RCW importing takes place before on-chip PLLs are configured.

The RCW is 512 bits long in order to contain all necessary configuration information for the chip. RCW data is read from external memory and written to the RCW status registers (see Reset control word status registers 1-16 (DCFG_RCWSR*n*)) contained in the Device Configuration and Pin Control module , after which the device is configured as specified in the RCW. Required format of data structure used by PBL provides details of the data structure that is required to reside in non-volatile memory.

4.6.4.1 RCW Field Definitions

The table below describes the function of the individual bits of the 512-bit (64-byte) RCW data structure.

NOTE

Unless noted otherwise, any bit ranges listed as reserved must be populated with 0.

Table 4-10. P3041 RCW Field Descriptions

Bit(s) (of 0-511)	Field Name	Description	Notes
PLL Configuration (Bits 0-127)			
0-1	SYS_PLL_CFG	System PLL Configuration	00 Platform frequency/system PLL targeting 667 MHz and above operation 01 Platform frequency/system PLL targeting below 667 MHz operation 10 Reserved 11 Reserved

Table continues on the next page...

Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
2-6	SYS_PLL_RAT	System PLL multiplier/ratio	This field selects the platform clock: SYSCLK ratio. 0000 Reserved 0100 4:1 0101 5:1 0110 6:1 0111 7:1 1000 8:1 1001 9:1 1010 10:1 1011 11:1 1100 12:1 **NOTE:** All PLL multipliers may not be supported .To check the PLL multipliers supported refer to the hardware specifications
7	Reserved		
8-9	MEM_PLL_CFG	Memory controller complex PLL configuration	This field configures the memory complex PLLs for the frequency of the reference clock applied. Note that in the case of synchronous mode operation, the reference clock to the memory complex PLLs is the platform clock (that is, the output of the system PLL). In the case of asynchronous mode operation, the reference clock is SYSCLK. See the hardware specifications for the definition of the boundary or cutoff point between the lower and higher reference clock frequency for each specific ratio/multiplier selected. Options: 00 Lower frequency reference clock 01 Higher frequency reference clock 10-11 Reserved
10-14	MEM_PLL_RAT	Memory controller complex PLL multiplier/ratio . Note that certain ratios are only valid in either synchronous mode or asynchronous mode. The DDR_SYNC RCW field is used to enable synchronous mode operation	The cutoff reference clock frequency for each option identifies the transition point when the MEM_PLL_CFG field must be configured to use the higher reference clock frequency option. Any reference clock frequency applied at or below the stated cutoff point requires configuring MEM_PLL_CFG for a lower frequency reference clock. Any reference clock frequency applied above the stated cutoff point requires configuring MEM_PLL_CFG for a higher frequency reference clock. See the hardware specifications for more details.
10-14 (con)	MEM_PLL_RAT (con)	(see row above)	0_0000 Reserved 0_0001 1:1 (sync mode) 600 MHz cutoff

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
			0_0010-0100 Reserved 0_0101 5:1 (async mode) 96.7 MHz cutoff 0_0110 6:1 (async mode) 80.6 MHz cutoff 0_0111 Reserved 0_1000 8:1 (async mode) 120.9 MHz cutoff 0_1001 9:1 (async mode) 107.4 MHz cutoff 0_1010 10:1 (async mode) 96.7 MHz cutoff 0_1011 Reserved 0_1100 12:1 (async mode) 80.6 MHz cutoff 0_1101 13:1 (async mode) 74.4 MHz cutoff 0_1110-0_1111 Reserved 1_0000 16:1 (async mode) 60 MHz cutoff 1_0001 Reserved 1_0010 18:1 (async mode) 53.7 MHz cutoff 1_0011 19:1 (async mode) 50.9 MHz cutoff 1_0100 20:1 (async mode)48.4 MHz cutoff 1_0101-1_0111 Reserved 1_1000 24:1 (async mode) 40.3 MHz cutoff Not all ratios are supported due to frequency restrictions. See the hardware specifications for more details.
15-63	Reserved		
64-65	CC1_PLL_CFG	Core cluster PLL 1 configuration.	00 Core cluster PLL output frequency targeting 1 GHz and above 01 Core cluster PLL output frequency targeting below 1 GHz 10-11 Reserved
66-70	CC1_PLL_RAT	Core cluster PLL 1 multiplier/ratio.	0_0000 Reserved 0_1000 8:1 Async 0_1001 9:1 Async 0_1010 10:1 Async 0_1011 11:1 Async 0_1100 12:1 Async 0_1101 13:1 Async 0_1110 14:1 Async 0_1111 15:1 Async 1_000 16:1 Async

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
			1_0001 Reserved 1_0010 18:1 Async 1_0011 Reserved 1_0100 20:1 Async 1_0101 Reserved 1_0110 22:1 Async 1_0111 Reserved 1_1000 Reserved
71	Reserved		
72-73	CC2_PLL_CFG	Core cluster PLL 2 configuration.	00 Core cluster PLL output frequency targeting 1 GHz and above 01 Core cluster PLL output frequency targeting below 1 GHz 10-11 Reserved
74-78	CC2_PLL_RAT	Core cluster PLL 2 multiplier/ratio.	0_0000 Reserved 0_1000 8:1 Async 0_1001 9:1 Async 0_1010 10:1 Async 0_1011 11:1 Async 0_1100 12:1 Async 0_1101 13:1 Async 0_1110 14:1 Async 0_1111 15:1 Async 1_000 16:1 Async 1_0001 Reserved 1_0010 18:1 Async 1_0011 Reserved 1_0100 20:1 Async 1_0101 Reserved 1_0110 22:1 Async 1_0111 Reserved 1_1000 Reserved
79-95	Reserved		
96-99	C0_PLL_SEL	Core complex 0 PLL select. Selects one of two available core complex PLLs and one of the two available VCO dividers of that PLL for the given core.	0000 CC1 PLL/1 0001 CC1 PLL/2 0010 CC1 PLL/4 0100 CC2 PLL/1

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
			0101 CC2 PLL/2 0110 CC2 PLL /4
100-103	C1_PLL_SEL	Core complex 1 PLL select. Selects one of two available core complex PLLs and one of the two available VCO dividers of that PLL for the given core.	0000 CC1 PLL/1 0001 CC1 PLL/2 0010 CC1 PLL/4 0100 CC2 PLL/1 0101 CC2 PLL/2 0110 CC2 PLL /4
104-107	C2_PLL_SEL	Core complex 2 PLL select. Selects one of two available core complex PLLs and one of the two available VCO dividers of that PLL for the given core.	0000 CC1 PLL/1 0001 CC1 PLL/2 0010 CC1 PLL/4 0100 CC2 PLL/1 0101 CC2 PLL/2 0110 CC2 PLL /4
108-111	C3_PLL_SEL	Core complex 3 PLL select. Selects one of two available core complex PLLs and one of the two available VCO dividers of that PLL for the given core.	0000 CC1 PLL/1 0001 CC1 PLL/2 0010 CC1 PLL/4 0100 CC2 PLL/1 0101 CC2 PLL/2 0110 CC2 PLL /4
112-127	Reserved		
SerDes PLL and Protocol Configuration (Bits 128-183)			
128-133	SRDS_PRTCL	SerDes protocol select	64 configurations are available. See SerDes Lane Assignments and Multiplexing, for a complete list of the options and the definitions of this encoded field. See Reference Clocks for SerDes Protocols, for a list of the legal combinations of reference clocks and SRDS_DIV_B*n* and SRDS_RATIO_B*n* encodings for each given SRDS_PRTCL encoding.
134	Reserved		
135	SRDS_RIO_SPD	SerDes RapidIO speed	Configures the serial RapidIO controllers associated with Bank 1 to be running at 3.125 Gbps/lane as opposed to either 2.5 or 5 Gbps/lane. 0 Serial RapidIO controllers running at either 2.5 or 5 Gbps/lane 1 Serial RapidIO controllers running at 3.125 Gbps/lane

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
			NOTE: This bit is only relevant to certain SerDes options where 2.5/3.125/5-G RapidIO and possibly debug are enabled. In addition, note that the bit rate of any debug lanes enabled in bank1 for the given option also runs at the same bit rate as the RapidIO controllers.
136-138	SRDS_RATIO_B1	SerDes PLL ratio-bank 1	Configures the PLL multiplier for the bank 1 PLL. 000 Reserved 001 Reserved 010 25:1 011 40:1 100 50:1
139-143	SRDS_DIV_B1	SerDes PLL dividers-bank 1	Configures the post-divider off of the bank 1 PLL's output. There is one bit for each of the five pairs of lanes in bank 1. The first bit corresponds to lanes A/B, while the last bit corresponds to lanes I/J. 0 Divide by 1 off of bank 1 PLL 1 Divide by 2 off of bank 1 PLL **NOTE:** When SerDes bank 1 is configured for 2.5- or 5-G operation (by SRDS_RATIO_B1 and the reference clock), lanes configured as SGMII operate at 1.25 Gbps regardless of the corresponding SRDS_DIV_B1 settings.
144-146	SRDS_RATIO_B2	SerDes PLL ratio-bank 2	Configures the PLL multiplier for the bank 2 PLL. 000 Reserved 001 20:1 010 25:1 011 40:1 100 50:1 101-111 Reserved
147	SRDS_DIV_B2	SerDes PLL divider-bank 2	Configures the post-divider off of the bank 2 PLL's output. 0 Divide by 1 off of bank 2 PLL 1 Divide by 2 off of bank 2 PLL **NOTE:** When SerDes bank 2 is configured for 2.5- or 5-GHz operation (by SRDS_RATIO_B2 and the reference clock), lanes configured as SGMII operate at 1.25 Gbps regardless

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
			of the corresponding SRDS_DIV_B2 settings.
148-150	SRDS_RATIO_B3	SerDes PLL ratio-bank 3	Configures the PLL multiplier for the bank 3 PLL. 000 Reserved 001 20:1 010 25:1 011 40:1 100 50:1 101 24:1 110 30:1 111 Reserved
151	SRDS_DIV_B3	SerDes PLL divider-bank 3	Configures the post-divider off of the bank 3 PLL's output. 0 Divide by 1 off of bank 3 PLL 1 Divide by 2 off of bank 3 PLL Note: When SerDes bank1 is conÞgured for 2.5G or 5G operation (by SRDS_RATIO_B3 given an applied reference clock frequency), lanes conÞgured as SGMII operate at 1.25Gbps regardless of the corresponding SRDS_DIV_B3 settings.
152-161	SRDS_LPD_B1	SerDes lane power down-bank 1	One bit per lane for the 10lanes in bank 1. 0 Lane not powered down 1 Lane powered down
162-165	SRDS_LPD_B2	SerDes lane power down-bank 2	One bit per lane for the 4 lanes in bank 2. 0 Lane not powered down 1 Lane powered down
166-169	SRDS_LPD_B3	SerDes lane power down-bank 3	One bit per lane for the 4 lanes in bank 3. 0 Lane not powered down 1 Lane powered down **NOTE:** When the selected SerDes option is defined such that any bank 3 lanes are unavailable for use, the corresponding bit(s) in this field must be set (lane(s) powered down) to ensure correct operation.
170-177	Reserved		

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
178	SRDS_EN	SerDes enable	**NOTE:** If the SerDes is disabled, the SerDes PLL locking time is skipped over during the hard reset or PORESET sequence. 0 SerDes disabled 1 SerDes enabled
179-183	Reserved		
184	DDR_SYNC	DDR synchronous mode.	Options: 0 memory controller complex run in asynchronous mode. In this mode, the reference clock to the memory controller complex PLL is SYSCLK. 1 memory controller complex run in synchronous mode. In this mode, the reference clock to the memory controller complex PLL is the platform clock.
185-191	Reserved		
Boot Configuration (Bits 192-223)			
192-195	PBI_SRC	Pre-boot initialization source. The pre-boot loader fetches address/ data pairs from the selected interface for the purpose of pre-boot initialization of CCSR and/or local memory space.	The following restrictions apply: 1) RCW and pre-boot initialization data must be loaded from the same non-volatile memory device. 2) At most, only one given eLBC option can be used for RCW, PBI_SRC, and BOOT_LOC. This does not mean that BOOT_LOC cannot be different from RCW and PBI_SRC (provided restriction 2 is observed). For example, the device can be configured to load RCW and pre-boot initialization data from NAND Flash (eLBC FCM) and boot from DDR. However, the device cannot load RCW and pre-boot initialization data from NAND Flash (eLBC FCM) and boot from NOR Flash (eLBC GPCM). Also, the hard-coded RCW source options are not considered their own memory interface for this purpose. 0000 I^2C1 normal addressing (up to 256 byte ROMs) 0001 I^2C1 extended addressing 0010 reserved 0011 reserved 0100 SPI 16-bit addressing 0101 SPI 24-bit addressing 0110 eSDHC 1000 eLBC FCM 8-bit small page (512-byte page) NAND Flash 1001 eLBC FCM 8-bit large page (2-Kbyte page) NAND Flash 1010 - reserved

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
			1011 - reserved 1100 eLBC GPCM 8-bit NOR Flash 1101 eLBC GPCM 16-bit NOR Flash 1110 eLBC GPCM 32-bit NOR Flash 1111 Disabled
196-200	BOOT_LOC	Boot location	Note the restrictions in PBI_SRC above. Specifically, restriction 2 that at most, only one given eLBC option can be used/configured as the source of CFG_RCW_SRC, PBI_SRC, and BOOT_LOC. 0_0000 PCI Express 1 0_0001 PCI Express 2 0_0010 PCI Express 3 0_0011 PCI Express 4 0_1000 sRIO1 0_1001 sRIO2 0_1110 - reserved 0_1111 - reserved 1_0000 Memory complex 1 1_1000 eLBC FCM 8-bit small page (512-byte) NAND Flash 1_1001 eLBC FCM 8-bit large page (2-Kbyte page) NAND Flash 1_1100 eLBC GPCM 8-bit NOR Flash 1_1101 eLBC GPCM 16-bit NOR Flash 1_1110 eLBC GPCM 32-bit NOR Flash 1_1111 Reserved **NOTE:** When the Boot location is selected to be any PCI Express controller configured in RC mode, the boot image is stored in the memory of some PCI Express downstream EP device. Such boot options require all the related PCI Express downstream devices involved in booting to be able to power up ahead of the PCI Express RC system and configure themselves before the boot code fetching. In other words, when the PCI Express RC system powers up later in such boot options, the first transaction the PCI Express RC controller performs is an outbound memory read transaction to fetch the boot code. It is unable to issue configuration transaction to configure and pre-initialize all the PCI Express downstream

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
			devices involved in booting. These boot options also require some nonvolatile memory to hold the necessary PBI code for LAW and PCI Express outbound ATMU window configuration purpose such that the PCI Express outbound memory read transaction can be sent out for boot code fetching.
201	BOOT_HO	Boot holdoff mode	0 All cores except core 0 in holdoff 1 All cores in holdoff
202	SB_EN	Secure boot enable	Note that secure boot is enabled if either this RCW bit is set or the Intent to secure fuse value is set. See Secure Boot and Trust Architecture, for more information. 0 Secure boot is not enabled 1 Secure boot is enabled-boot from internal boot ROM
203-223	Reserved		
Clocking Configuration (Bits 224-255)			
224	Reserved		
225	FM_CLK_SEL	Frame manager clock select	Allows for Fman frequency to be maximized (using async clock) if Fman component is present and async clock source is provided. 0 Platform clock /2 1 Core cluster PLL 2 /2 Fman runs asynchronous with respect to platform.
226-228	Reserved		
229	HWA_ASYNC_DIV	Hardware accelerator async divider	Select frequency ratio for the accelerator block (Fman) operating asynchronous with respect to platform. 0 Core cluster PLL 2 /2 1 Core cluster PLL 2 /4
230-231	DRAM_LAT	DDR latency.	Options: 00 6-6-6 or 7-7-7 DRAMs 01 8-8-8, 9-9-9, 10-10-10, or 11-11-11 DRAMs 10 Reserved 11 5-5-5 DRAMs **NOTE:** If DDR latency is not known at the time the RCW is loaded, it is acceptable to conservatively configure this field as 01. This field is used for optimizing the DDR interface. No functional issues result if this

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
			field does not match the latency of the actual DRAM's used.
232	DDR_RATE	DDR data rate.	Refer to the hardware specifications for settings.
233	Reserved		
234	DDR_RSV0	Reserved	Refer to the hardware specifications for settings.
235-255	Reserved		
Memory and High-Speed I/O Configuration (Bits 256-287)			
256-259	Reserved		
260-262	RIO_DEVICE_ID	RapidIO device ID	See description of BDID field in the BDIDCSR register for how this field is used by the SRIO block.
263	RIO_SYS_SIZE	RapidIO system size	See description of CTLS field within the PEFCAR register for how this field is used by the SRIO block. 0 Small system size (256 devices) 1 Large system size (65,536 devices)
264-266	HOST_AGT_B1	Host/agent bank 1. Configures host (root complex) or agent (endpoint) mode for selected serial RapidIO/PCI Express interfaces on SerDes Bank 1, lanes A-H	000 All host mode 001 All agent mode 010 First two agent mode 011 First and third agent mode 100 First and fourth agent mode 101 Second and third agent mode 110 Second and fourth agent mode 111 Third and fourth agent mode For these options, "first" refers to the PCI Express or serial RapidIO controller assigned to the lanes closest to lane A, as shown in SerDes Lane Assignments and Multiplexing for the given SerDes Bank 1 configuration; "second" refers to the next PCI Express or serial RapidIO controller assigned to the lanes to the right of the "first," and so on. **NOTE:** Any dual serial RapidIO configurations must have both interfaces defined as host or both interfaces defined as agent. Results are undefined if an illegal host/agent configuration is used. **NOTE:** The HOST_AGT_B1 does not control the host/agent mode configuration of controllers occupying SerDes Bank 1, lane I and J. When there are fewer than four controllers used for a specific SRDS_PRTCL selection, only those applicable controllers in Bank 1 lane A to H are controlled by the above HOST_AGT_B1 options. For example, if SRDS_PRTCL is set to 0x02 while

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
			HOST_AGT_B1 is set to 101, since there are only two PCI Express controllers (PCIe1 and PCIe2) used in Bank 1 lane A to H, the "third" controller mentioned in this HOST_AGT_B1 option is irrelevant. In other words, it results in the configuration to be: PCIe1 as host and PCIe2 as agent.
267	HOST_AGT_B2	Host/agent bank 2(and bank 1 lane I when used by PCI Express). Configures root complex/endpoint mode for selected PCI Express interface on SerDes bank 2and bank 1 lane I.	0 Host (root complex) 1 Agent (endpoint)
268	HOST_AGT_B3	Host/agent bank 3. Configures root complex/endpoint mode for selected serial RapiodIO interface on SerDes Bank 3.	0 Host (root complex) 1 Agent (endpoint) **NOTE:** Any dual serial RapidIO configurations must have both interfaces defined as host or both interfaces defined as agent. Results are undefined if an illegal host/agent configuration is used.
269-287	Reserved		
General Purpose Information (Bits 288-319)			
288-319	GP_INFO	General purpose information. This field has no effect on functional logic; it may be used by software.	
320-351	Reserved		
Pin Multiplexing Configuration (Bits 352-415)			
Group A Pin Configuration (Bits 352-383)			
352-353	1588	These bits determine whether GPIO or 1588 functionality is selected for the given pin.	0 1588 functionality-TSEC_1588_ALARM_OUT2, TSEC_1588_PULSE_OUT2 1 GPIO functionality-GPIO[30:31]
354	Reserved		
355-357	I^2C	Configures the functionality of the I^2C3 and I^2C4 pins.	000 {IIC3_SCL, IIC3_SDA, IIC4_SCL, IIC4_SDA} 001 {IIC3_SCL, IIC3_SDA, EVT[5:6]}_B 010 {GPIO[16:17], IIC4_SCL, IIC4_SDA} 011 {GPIO[16:17], EVT[5:6]_B} 100 {SDHC_CD_B, SDHC_WP, IIC4_SCL, IIC4_SDA} 101 {SDHC_CD_B, SDHC_WP, EVT[5:6]_B} 110-111 Reserved **NOTE:** EVT[5:6] functionality, if selected by this field, defaults to input functionality until reconfigured by software to output functionality.
358-359	Reserved		

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
360-361	EC1	This configures the functionality assigned to the EC1 parallel mode pins.	00 Frame ManagerMAC4 RGMII 01 Reserved 10 Reserved 11 No parallel mode Ethernet **NOTE:** If SGMII mode is to be used (with appropriate SerDes lane powered "on" via SRDS_LPD*n* setting) for MAC4, then RGMII mode must not be enabled via this EC1 setting. However, if RGMII mode is to be used by MAC4 and EC1 is set accordingly, then SGMII mode must not be enabled for MAC4 (appropriate SerDes lane must be powered off via SRDS_LPDn setting if SGMII is selected for that bank *n* via SRDS_PRTCL setting). **NOTE:** . **NOTE:**
362	Reserved		
363-364	EC2	This configures the functionality assigned to the EC2 parallel mode pins.	00 Frame Manager MAC5 RGMII 01 Reserved 10 Reserved 11 No parallel mode Ethernet **NOTE:** If SGMII mode is to be used (with appropriate SerDes lane powered "on" via SRDS_LPD*n* setting) for MAC5, then RGMII mode must not be enabled via this EC2 setting. However, if RGMII mode is to be used by MAC5 and EC2 is set accordingly, then SGMII mode must not be enabled for MAC5 (appropriate SerDes lane must be powered off via SRDS_LPDBn setting if SGMII is selected for that bank *n* via SRDS_PRTCL setting). **NOTE:**
365	GPIO	Configures functionality of GPIO[4:7] pins.	0 {GPIO[4:7]} 1 {USB1_DRVVBUS, USB1_PWRFAULT, USB2_DRVVBUS, USB2_PWRFAULT}
366-368	UART	Configures functionality of the UART pins, as follows: UART1_SOUT, UART1_SIN, UART1_RTS_B, UART1_CTS_B, UART2_SOUT, UART2_SIN, UART2_RTS_B, UART2_CTS_B	000 {GPIO[8], GPIO[10], GPIO[12], GPIO[14], GPIO[9], GPIO[11], GPIO[13], GPIO[15]} 001-010 Reserved 011 {UART1_SOUT, UART1_SIN, GPIO[12], GPIO[14], GPIO[9], GPIO[11], GPIO[13], GPIO[15]} 100 {UART1_SOUT, UART1_SIN, UART1_RTS_B, UART1_CTS_B, GPIO[9], GPIO[11], GPIO[13], GPIO[15]}

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
			101 {UART1_SOUT, UART1_SIN, GPIO[12], GPIO[14], UART2_SOUT, UART2_SIN, GPIO[13], GPIO[15]} 110 {UART1_SOUT, UART1_SIN, UART1_RTS_B, UART1_CTS_B, UART2_SOUT, UART2_SIN, UART2_RTS_B, UART2_CTS_B} 111 {UART1_SOUT, UART1_SIN, UART3_SOUT, UART3_SIN, UART2_SOUT, UART2_SIN, UART4_SOUT, UART4_SIN}
369	Reserved		
370	SPI	Configures functionality of the SPI_CS[0:3] pins.	0 SPI_CS[0:3]_B 1 SDHC_DAT[4:7] for 8-bit MCC card support
371	Reserved		
372-374	IRQ1	Configures functionality of IRQ[3:5] pins-the corresponding GPIOs for these pins are GPIO[21:23]	0 IRQ 1 GPIO
375	Reserved		
376-379	IRQ2	Configures functionality of IRQ[6:9] pins-the corresponding GPIOs for these pins are GPIO[24:27].	0 IRQ 1 GPIO
380	Reserved		
381-382	IRQ3	Configures functionality of IRQ[10:11] pins-the corresponding GPIOs for these pins are GPIO[28:29].	0 IRQ 1 GPIO
383	IRQ_OUT	Configures functionality of the IRQ_OUT pin.	0 IRQ_OUT_B 1 EVT[9]_B **NOTE:** EVT[9]_B functionality, if selected by this field, defaults to input functionality until software reconfigures to output functionality.
Group B Pin Configuration (Bits 384-415)			
384	DMA1	Configures functionality of the DMA1 pins.	0 {DMA1_DREQ0_B, DMA1_DACK0_B, DMA1_DDONE0_B} 1 {GPIO[18:19], Reserved }
385	Reserved		
386-387	DMA2	Configures functionality of the DMA2 pins.	00 {DMA2_DREQ0_B, DMA2_DACK0_B, DMA2_DDONE0_B} 01 {LB_DVAL, LB_SRCID[0:1]} 10 {GPIO[20], EVT[7]_B, EVT[8]_B} EVT[7:8]_B functionality, if selected by this field, defaults to input functionality until reconfigured by software to output functionality.

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Table 4-10. P3041 RCW Field Descriptions (continued)

Bit(s) (of 0-511)	Field Name	Description	Notes
			11 Reserved Note: EVT_[7:8] Bfunctionality, if selected by this Þeld, defaults to input functionality until reconÞgured by software to output functionality
388	GTX_CLK125_SEL	Selects unique/dedicated GTX_CLK125 reference clcok sources as opposed to common GTX_CLK125 reference clock source for the two dTSEC blocks associated with the EC1_xxx and EC2_xxx RGMII pins.	0 EC1_GTX_CLK125 is the RGMII reference clock source for both dTSEC4 and dTSEC5 1 EC1_GTX_CLK125 is the RGMII reference clock source for dTSEC4 and EC2_GTX_CLK125 is the RGMII reference clock source for dTSEC5
389-511	Reserved		

4.6.4.1.1 Hard Coded RCW Options

The device can be initialized with a hard-coded RCW option, selected by the cfg_rcw_src[0:4] configuration inputs. When the RCW source is set to a hard-coded RCW option, the device is automatically configured according to the specific RCW field encodings pre-assigned for the given option.

All hard-coded RCW options allow the reset sequence to bypass using the PBL to load in RCW data from a non-volatile memory device. One envisioned use of the hard-coded RCW options is to facilitate booting with enough basic functionality to allow reprogramming of a corrupted (or perhaps uninitialized) non-volatile memory device that originally contained RCW information.

Table 4-11 describes hard-coded RCW configuration options. All configuration options listed also define the following characteristics:

- Interfaces or subsets of signals that offer multiplexed functionality are not driven (that is, high-impedance) or disabled. This means that external interrupt (IRQ*n*), GPIO, external DMA, I^2C3, I^2C4, and IRQ_OUT signals are not available. Exceptions are:
- UART signals
- IRQ[0:2]
- GPIO[0:7] except GPIO[4:7] is configured for USB PWRFAULT and DRVVBUS
- Unused SerDes lanes are powered down.